TEV protease production and cleavage

The 27 kDa catalytic domain of the viral NIa polyprotein of Tobacco Etch Virus (TEV) is a cysteine protease, with a Cys-Asp-His catalytic triad, that cleaves `ENLYFQ/G` (or `ENLYFQ/S`) between the P1 Gln and the P1′ Gly/Ser. The autolysis-resistant S219V variant (Kapust et al., 2001) is about 100-fold more stable against self-cleavage than the wild-type protease, and a slightly better catalyst. Its narrow specificity, broad activity window (pH 6.0 to pH 9.0, 4–37 °C), and tolerance of common purification buffers make TEV the standard choice for removing affinity tags downstream of His-, MBP-, or GST-capture.

The construct is suitable for cleavage in solution. For on-column cleavage, PreScission (3C, `LEVLFQ/GP`) or SUMO (Ulp1) SOP 0048 are preferred.

A single 1 L autoinduction prep yields tens of milligrams of protease, enough to cleave hundreds of milligrams of tagged target, at a fraction of the cost of commercial protease.

Risk assessment

▷ Wear gloves, safety glasses, lab coat



Reviewed: Oct 30, 2026

Procedures

›› *Preparative expression*

- ☐ Erlenmeyer flask, 1–2 L, baffled
  - *or*: Fermenter, 4 L capacity
- ☐ Centrifuge bottle, 1 L
- ☐ TB-AI ◇ R0205
- ☐ *E. coli* BL21(DE3)
- ☐ pOPIN-His6-MBP-(3C)-TEV, 70.3 kDa
  - *or*: pRK793, MBP-(TEV)-His7-TEV-Arg5, 28.6 kDa, Addgene
  - *or*: pRK1043, MBP-TEV-Arg5, 70.0 kDa, Addgene

(1.) Inoculate 50 mL LB containing carbenicillin at 100 mg/L with a single fresh colony of *E. coli* BL21(DE3) carrying the TEV expression plasmid. Incubate overnight at 37 °C, 180 rpm.

> ***Hint:*** All three plasmids listed carry the β-lactamase marker. Carbenicillin is stable for 18–24 h at room temperature, while secreted β-lactamase strips ampicillin, allowing plasmid-free cells to overtake the culture. Transform freshly SOP 0034 rather than working from a glycerol stock. Colonies older than two or three days accumulate non-expressing variants.

(2.) Retain the equivalent of 0.5 OD mL of cells from the pre-culture in 70 µL 1 × SDS-PAGE sample buffer SOP 0045 as the uninduced reference; the autoinduction medium has no uninduced phase.

(3.) Inoculate 1 L TB autoinduction medium with 10 mL of the overnight pre-culture in a baffled Erlenmeyer flask. Incubate at room temperature (20–25 °C), 180 rpm, for 18–24 h, typically overnight.

(4.) Retain a second sample of the same cell equivalent at harvest density.

(5.) Harvest the cells at 4 000 × *g* for 20 min at 4 °C. Discard the supernatant and weigh the wet pellet.

> ***Hint:*** Pellets may be stored at −80 °C for months before lysis.

›› *Lysis*

- ☐ TEV lysis buffer, pH 8.0 ◇ R0206
- ☐ 100 mM AEBSF ◇ R0213
- ☐ DNase I, 10 U/µL stock
- ☐ 1 M $MgCl_2$ ◇ R0031

(1.) Resuspend the pellet in 5 mL cold TEV lysis buffer per 1 g wet mass. Supplement AEBSF to 0.5 mM, $MgCl_2$ to 2 mM, and DNase I to 40 U/mL final. Keep the lysate on ice.

> ***This is why:*** DNase I is Mg^{2+}-dependent, so TEV lysis buffer must be spiked. The magnesium is removed on the column and does not reach the protease product.

(2.) Lyse the cells by sonication on ice (0.125 in horn; 6 cycles of 30 s on, 60 s off, at 30–40% amplitude) or by two passes through a high-pressure homogenizer.

> ***Hint:*** Final lysate volume from 1 L TB-AI is typically 50–75 mL.

(3.) Clear the lysate by centrifugation at 35 000–50 000 × *g* for 20 min at 4 °C.

(4.) Retain 20 µL of the cleared lysate, and resuspend a comparable aliquot of the pellet in 1 × SDS-PAGE sample buffer as the insoluble fraction.

›› *Affinity chromatography (HisTrap FF on ÄKTA)*

- ☐ FPLC system
- ☐ HisTrap FF (crude) column, 5 mL capacity
- ☐ TEV elution buffer, pH 8.0 ◇ R0207

(1.) Filter the cleared lysate through a 0.45 µm filter immediately before loading.

> ***Critical:*** An unfiltered lysate clogs the HisTrap FF crude inlet within minutes at the loading flow rate; do not skip filtration.

(2.) Run the FPLC variant, route *Alternative C* of SOP 0045, on the 5 mL HisTrap FF crude column, with TEV lysis buffer as the binding buffer and TEV elution buffer as the eluent in place of the phosphate-based pair given there. The load is the 50–75 mL of cleared, filtered lysate from 1 L of culture. Where no FPLC is available, run route *Alternative A* by gravity instead.

> ***Hint:*** Skip the imidazole spike before loading; TEV lysis buffer already carries 20 mM imidazole.

> ***Hint:*** The His6-MBP-TEV peak elutes early in the shallow 10–50% segment of the gradient, and tails across it. No window is recorded for a His7-TEV-Arg5 lot.

(3.) Analyze the peak fractions by SDS-PAGE SOP 0007 and inspect the FPLC chromatogram. Pool the fractions carrying a single sharp band at 70 kDa (His6-MBP-TEV) or 28 kDa (His7-TEV-Arg5), taking the peak proper and leaving a shoulder on either side of it out rather than pooling to baseline.

> ***Critical:*** The 3C linker in pOPIN is not a TEV substrate, so the fusion protein stays intact. pRK793 self-cleaves instead: the freed MBP (43 kDa) carries no His tag and washes through, but fusion that failed to cleave (72 kDa) is tagged and does bind.

›› *Dialysis and concentration*

- ☐ 5 × TEV storage buffer, pH 8.0 ◇ R0208
- ☐ Dialysis membrane tube, MWCO 12–14 kDa
- ☐ 500 mM TCEP ◇ R0215

(1.) Add TCEP to 0.5 mM to the pooled fractions. Pre-rinse the dialysis tubing and transfer the material into it.

(2.) Dialyze overnight at 4 °C against 1 × dialysis buffer at a sample-to-buffer ratio of 1 : 25 or higher.

> ***Critical:*** 1.5 L dialysis buffer brings a sample of up to 60 mL below 20 mM imidazole. Split larger pools across two baths.

(3.) Recover the dialysate. The in-bag TCEP has dialysed out by now, so add TCEP again to 0.5 mM final. Clear the dialysate at 20 000 × *g* for 15 min at 4 °C to remove any precipitate.

(4.) Determine the concentration spectrophotometrically at 280 nm. Bring to 2 g/L with 1 × dialysis buffer. If below 2 g/L, concentrate over an ultrafiltration device (SOP 0045).

> ***Critical:*** Do not concentrate past 2.0 g/L. The protease precipitates against the membrane.

> ***Hint:*** His6-MBP-TEV has $\varepsilon_{280} = 98\,570\ \mathrm{M^{-1}\,cm^{-1}}$ and $a_{280} = 1.40\ \mathrm{L\,g^{-1}\,cm^{-1}}$. His7-TEV-Arg5 has $\varepsilon_{280} = 31\,970\ \mathrm{M^{-1}\,cm^{-1}}$ and $a_{280} = 1.12\ \mathrm{L\,g^{-1}\,cm^{-1}}$.

> ***Quality assurance:*** A typical His-TEV preparation yields on the order of 30 mg per 1 L of culture.

(5.) Add 1.0 vol glycerol diluent. Mix until homogeneous, do not shake.

> ***Hint:*** The protease will be about 1 g/L in 25 mM Tris, 500 mM NaCl, 50% glycerol (v/v), 0.5 mM TCEP; comparable to commercial preparations. At that concentration, the lot does not freeze when stored at −20 °C, so thawing costs no activity loss.

›› *Aliquot, storage, and lot record*

(1.) Aliquot the protease into 50–100 µL portions and store at −20 °C.

(2.) Record the following lot metadata in the protease lot log:

- Lot number as `MM/YYYY`, following the labelling convention in SOP 0030.
- Date of preparation and contributor.
- Final concentration (mg/mL) and total yield (mg).
- Number and size of aliquots.
- Specific activity (mg protease required to cleave 1 mg of 70 kDa substrate at 4 °C overnight).
- Storage buffer composition and storage location.

›› *Standard TEV activity assay*

☐ TEV assay buffer, pH 8.0 ◇ R0209	☐ TEV-cleavable test substrate, 68.8 kDa

(1.) Dilute the TEV-cleavable test substrate to 0.5 g/L with TEV assay buffer. Any purified His-MBP fusion carrying an `ENLYFQ/G` site will work. Activity is normalized to a nominal substrate mass of 70 kDa.

> ***Critical:*** One lot test consumes about 220 µL of the diluted substrate. The substrate must be purified before the test is started.

(2.) Thaw two separate TEV protease aliquots of the lot. From each, prepare 50 µL at 0.4 g/L in assay buffer. For a lot at 1 g/L this is 20 µL lot plus 30 µL buffer.

> ***Critical:*** The two aliquots carry the replication of the assay. Splitting one aliquot into two series replicates the pipetting but not the aliquot.

(3.) Carry each replicate down a five-point two-fold series in reaction tubes SOP 0031. Pre-fill tubes 2 to 5 with 25 µL assay buffer, transfer 25 µL from tube 1 into tube 2, mix, and continue with a fresh tip:

Tube	TEV concentration	TEV mass per assay	TEV per 1 mg substrate
1	0.400 g/L	2.000 µg	0.200 mg
2	0.200 g/L	1.000 µg	0.100 mg
3	0.100 g/L	0.500 µg	0.050 mg
4	0.050 g/L	0.250 µg	0.025 mg
5	0.025 g/L	0.125 µg	0.013 mg

(4.) Transfer 5 µL of each dilution of both series into fresh tubes. Add 20 µL of the 0.5 g/L substrate dilution (10 µg substrate; 25 µL total). Mix gently, spin down briefly, and incubate overnight at 4 °C.

(5.) Stop each reaction with 8 µL 4 × SDS-PAGE sample buffer SOP 0007 and heat to 95 °C for 5 min. Prepare substrate-only (20 µL substrate dilution plus 5 µL assay buffer) and protease-only (5 µL of tube 1 of either series plus 20 µL assay buffer) controls. ⏲ 18 h

> ***Critical:*** A His6-MBP-TEV lot and the uncleaved substrate differ by under 2 kDa and co-migrate on this gel; a His7-TEV-Arg5 lot at 28 kDa does not. Use the protease-only lane to place the protease, and read cleavage from the product bands rather than from the substrate position.

(6.) Resolve both series and the two controls on one 12% SDS-PAGE gel SOP 0007, loading 10 µL per lane. Stain with Coomassie SOP 0010. Look for the substrate band at 68.8 kDa to disappear in favor of the His-MBP (44 kDa) and Rab7 (23.7 kDa) product bands.

> ***Quality assurance:*** The twelve samples fit a 12-well comb; cast a 15-well gel to carry a molecular-weight marker alongside them. Keeping both series on one gel means the two endpoints are compared under identical staining.

⊞ Optional: *Using TEV protease for tag cleavage*

- ☐ Dialysis tubing or cassette
- ☐ Reverse IMAC column
- ☐ EDTA, 0.5 M stock, pH 8.0

(1.) Verify that the fusion places `ENLYFQ/G` between the tag and the target, with at least three residues of flexible linker between the cleavage site and the folded domain of the target. Cleavage leaves the Gly on the target N-terminus.

(2.) Match the cleavage mixture to the following reaction conditions:

- Buffering system: Tris, HEPES, phosphate, or MES buffer (pH 6.0 to pH 9.0).
- Temperature: 4–37 °C; activity peaks near 34 °C, so 37 °C costs little. The default, 4 °C overnight, is chosen for the target rather than for the protease.
- Reductant: 1 mM TCEP, or 1–5 mM DTT.
- Monovalent salt: fully active at low salt; half as active at 500 mM NaCl; keep the digest near 150 mM where the target allows.
- Imidazole: dilute or exchange an IMAC eluate to 20 mM or below before a long digest.
- Tolerated protease inhibitors: PMSF and AEBSF (1 mM), TLCK (1 mM), pepstatin A (1 mM), bestatin (1 g/L), E-64 (3 g/L), Roche "complete" cocktail (1 ×).

> ***Hint:*** Inhibitor figures are concentrations at which tolerance was demonstrated, not amounts to add. Each exceeds its usual working concentration. Standard inhibitor cocktails are tolerated.

> ***Hint:*** Two target classes override the reductant choice: for disulfide-bonded targets replace DTT with a redox buffer (3 mM reduced and 0.3 mM oxidized glutathione), for zinc-fingers replace DTT with β-mercaptoethanol and EDTA with citrate.

(3.) Avoid conditions incompatible with TEV protease activity:

- Zn^{2+} at 5 mM or above, Cu^{2+} at 1 mM or above, Co^{2+} at 10 mM or above. Add EDTA up to 1 mM to quench divalent metals.
- Glycerol or sorbitol exceeding 40% (w/v).
- Iodoacetamide and NEM are potent TEV protease inhibitors.
- For detergents, verify per detergent before scaling up; some are inhibitory.

(4.) Add TEV to the pooled chromatographic fractions at the ratio indicated by the current lot's activity assay (typically 0.05 mg TEV per 1 mg target). Take a zero-time sample for SDS-PAGE, mix with sample buffer, and freeze immediately.

> ***This is why:*** On-column cleavage is inefficient here: TEV binds nonspecifically to column materials; the His6-MBP-TEV also competes with the tagged substrate for the same Ni^{2+} sites, and the EDTA that protects TEV from divalent metals strips the resin. PreScission (3C) and SUMO (Ulp1) are the usual choices for on-column work.

(5.) Pre-rinse the dialysis tubing or cassette and transfer the cleavage mixture into it. Add TCEP to 0.5 mM directly into the dialysis vessel before sealing. Dialyze against the target's storage or next-step buffer at a sample-to-buffer ratio of at least 1 : 100 for 16 h at 4 °C. ⏲ 16 h

> ***Hint:*** Cleavage during dialysis removes imidazole and any low-molecular-weight inhibitors as the reaction runs.

(6.) Remove the His6-MBP-TEV protease and any released His-tagged portion of the substrate by reverse IMAC 🗎 SOP 0045. The cleaved, tag-free target appears in the flow-through.

> ***Critical:*** Remove EDTA and strong reductants such as DTT before reverse IMAC; EDTA strips Ni^{2+} from the resin and DTT reduces it. The preceding dialysis does this provided the dialysis buffer contains neither. TCEP at 1 mM, as used throughout this protocol, is tolerated on immobilized nickel and does not have to be removed.

(7.) Confirm complete cleavage by SDS-PAGE SOP 0007 against the zero-time sample. Expect loss of the tagged precursor band and gain of two product bands.

Analyses

- Resolve the retained samples (uninduced and induced culture, cleared lysate, insoluble fraction, column flow-through, and the pooled elution fractions) on one SDS-PAGE gel SOP 0007 and stain with Coomassie SOP 0010, loading 10 µL of each culture sample.

  Quality assurance: A usable lot shows the protease in the cleared lysate and concentrated in the elution fractions, with the flow-through and washes carrying the bulk of the host protein.

- Record the yield and the concentration against both the culture volume and the wet pellet weight taken at harvest. Expect on the order of 30 mg per 1 L of culture.

- Confirm identity by intact-mass LC-ESI-MS SOP 0045; accept within 5 Da.

  Quality assurance: A retained N-terminal methionine or gluconoylation at 178 Da are the contaminant shifts to expect.

- Record the lot's specific activity as the highest endpoint whose freed Rab7 band at 23.7 kDa is as strong as in the tubes below it. The first tube in which that band weakens marks the end of complete cleavage.

  Hint: The assay buffer carries 500 mM NaCl, where the protease is about half as active as at low salt, so a figure recorded here is conservative for a digest run nearer 150 mM. Reformulating the buffer would shift every recorded figure and orphan the earlier lots.

  Critical: Read the endpoint from the product, not from the loss of the precursor. A His6-MBP-TEV lot co-migrates with the uncleaved substrate at 68.8 kDa and covers it, so surviving substrate cannot be judged there. Of the two products, only Rab7 is unambiguous; the 44 kDa His-MBP band coincides with the 43 kDa MBP that a pRK793 lot frees by self-cleavage.

  Critical: The endpoint must fall inside the series. If every tube cleaves completely, the recorded figure is only the most dilute tube that was tried and not the lot's activity: extend the ladder by further twofold steps and re-run before recording anything.

  Critical: The two endpoints must agree within one tube. On a discrepancy of two tubes or more, run a third series from a third aliquot and report the median; do not average the three, because a two-fold ladder is an ordinal scale and the mean of two rungs is not a measured value.

- Reject the lot and repeat the preparation if the Rab7 band fails to reach full strength at the highest TEV concentration in either series. Otherwise release the lot and use its recorded specific activity to dose every subsequent cleavage reaction.

  Hint: A typical preparation will fully cleave at 0.05 mg TEV per 1 mg substrate. A lot that needs a higher tube is still usable; it simply consumes proportionally more protease per digest.

Troubleshooting

Using TEV protease for tag cleavage

In Step 1:

- Inefficient cleavage
  - Match the TEV cleavage site exactly: even conservative substitutions at P2 (Phe to Tyr) or P4 (Leu to Phe) reduce processing by about two orders of magnitude.

In Step 7:

- Cleavage stalls at 50% overnight
  - Confirm sample is within the TEV buffer windows above; in particular, check for residual imidazole above 20 mM and undeclared divalent metals.
  - Use an unopened aliquot; a tube stored at −80 °C and thawed once already has lost activity.
  - Increase the TEV-to-target ratio twofold and extend the incubation to 24–36 h at 4 °C, or warm to 16 °C if the target tolerates it.
- Target precipitates during cleavage
  - Tagged partners such as MBP, SUMO, and GST are solubilizing, so the free target may be less soluble. Keep the target concentration below 1 g/L, raise salt to 300–500 mM NaCl, add 10% glycerol before TEV addition, or drop the cleavage temperature to 4 °C if it was higher.

Materials

Preparative expression

- ☐ **Cultivation vessel**
  - **Erlenmeyer flask**, 1–2 L, baffled
  - *or* **Fermenter**, 4 L capacity
- ☐ **Centrifuge bottle**, 1 L
- ☐ **TB autoinduction medium (TB-AI)** ◇ R0205
  Store at 4 °C.
- ☐ ***E. coli* BL21(DE3)**
- ☐ **TEV (S219V) expression plasmid**
  - **pOPIN-His6-MBP-(3C)-TEV**, 70.3 kDa
    This is why: His6-MBP is joined by a 3C linker (`LEVLFQ/GP`), which TEV cannot cleave, so the fusion survives its own protease. It has ε_{280} = 98 570 $M^{-1}\,cm^{-1}$.
  - *or* **pRK793**, MBP-(TEV)-His7-TEV-Arg5, 28.6 kDa
    RRID Addgene_8827
    This is why: Self-cleaves in the cell to release His7-TEV-Arg5, which has ε_{280} = 31 970 $M^{-1}\,cm^{-1}$.
  - *or* **pRK1043**, MBP-TEV-Arg5, 70.0 kDa
    RRID Addgene_8835
    Critical: This construct does not bind the nickel column used here, nor does it self-cleave. Capture on amylose. ***This is why:*** The fusion has ε_{280} = 98 320 $M^{-1}\,cm^{-1}$.

Lysis

- ☐ **TEV lysis buffer**, pH 8.0 ◇ R0206
  Store at 4 °C.
- ☐ **4-(2-Aminoethyl)benzenesulfonyl fluoride hydrochloride (AEBSF)**, 100 mM ◇ R0213
- ☐ **DNase I**, 10 U/µL stock
- ☐ **Magnesium chloride ($MgCl_2$)**, 1 M ◇ R0031

Affinity chromatography (HisTrap FF on ÄKTA)

- ☐ **FPLC system**
  - *e.g.* **ÄKTA pure**
    Cytiva
- ☐ **HisTrap FF (crude) column**, 5 mL capacity
  - **HisTrap FF crude**
    Cytiva, 17-5286-01
- ☐ **TEV elution buffer**, pH 8.0 ◇ R0207
  Store at 4 °C.

Dialysis and concentration

- ☐ **TEV storage buffer**, 5 ×, pH 8.0 ◇ R0208
  Store at 4 °C.
- ☐ **Dialysis membrane tube**, MWCO 12–14 kDa
  - *e.g.* **Spectra/Por® 4**
    Spectrum Labs, Spectra/Por® dialysis tubing
- ☐ **Tris(2-carboxyethyl)phosphine hydrochloride (TCEP)**, 500 mM ◇ R0215

Standard TEV activity assay

- ☐ **TEV assay buffer**, pH 8.0 ◇ R0209
  Store at 4 °C.
- ☐ **TEV-cleavable test substrate**, 68.8 kDa
  This is why: A His-MBP-(TEV)-Rab7 fusion. On cleavage it releases free Rab7 (23.7 kDa) and free His-MBP (44 kDa), both readily resolved on a 12% SDS-PAGE gel. The intact fusion has ε_{280} = 95 020 $M^{-1}\,cm^{-1}$.

Using TEV protease for tag cleavage

- ☐ **Dialysis tubing or cassette**
  Hint: Choose a MWCO below the mass of the freed target and well below the protease mass; 12–14 kDa is used here.
  - **Dialysis membrane tube**
  - **Dialysis cassette**
- ☐ **Reverse IMAC column**
  Hint: For the reverse-IMAC pass that removes the cleaved tag and the protease.
  - **HisTrap or Ni-NTA column**
- ☐ **EDTA**, 0.5 M stock, pH 8.0

Recipes

TB autoinduction medium (TB-AI)

Amount	Ingredient		Stock	Final
2 mL	Magnesium sulfate ($MgSO_4$) [10034-99-8]	☐	1 M	2 mM
2 mL	Carbenicillin (Crb) Skin sensitization (H317)	☐ R0013	50 g/L	100 mg/L
500 µL	Lactose	☐ [63-42-3]	20%	0.01%
To 1 L	Terrific broth (TB) medium, autoclaved			

Add magnesium sulfate, carbenicillin and lactose immediately before use. Store at 4 °C. ***This is why:*** TB carries no glucose, and neither its glycerol nor the amino acids represses the lac promoter. The lactose is therefore imported by the *lacY* permease and converted by *lacZ* β-galactosidase to allolactose, which releases LacI from the T7 polymerase gene. Even a small amount of lactose induces the culture. Expression starts shortly after inoculation.

TB autoinduction medium (TB-AI)

☐ 2 mM $MgSO_4$, ☐ 100 mg/L Crb, ☐ 0.01% Lactose



WARNING



Skin sensitization

☐ Hold for EHS review before disposal

Date: Sign: R0205

TEV lysis buffer, pH 8.0

Amount	Ingredient		Stock	Final
25 mL	Tris hydrochloride (Tris-Cl), pH 8.0 R0056		1 M	50 mM
30 mL	Sodium chloride (NaCl)	R0046	5 M	300 mM
1.67 mL	Imidazole Skin corrosion (H314); Reproductive toxicity (H360)	[288-32-4] R0027	6 M	20 mM
1 mL	TCEP Skin corrosion (H314)	☐ R0215	500 mM	1 mM
To 500 mL	Water, reagent-grade			

Adjust to pH 8.0 with sodium hydroxide. Add TCEP immediately before use. Store at 4 °C.

TEV lysis buffer

50 mM Tris-Cl, 300 mM NaCl, 20 mM Imidazole, ☐ 1 mM TCEP, pH 8.0



DANGER



Reproductive toxicity

☐ Collect as HAZARDOUS WASTE
☐ DO NOT wash into sewer

Date: Sign: R0206

TEV elution buffer, pH 8.0

Amount	Ingredient		Stock	Final
10 mL	Tris hydrochloride (Tris-Cl), pH 8.0 R0056		1 M	50 mM
12 mL	Sodium chloride (NaCl)	R0046	5 M	300 mM
16.7 mL	Imidazole Skin corrosion (H314); Reproductive toxicity (H360)	[288-32-4] R0027	6 M	500 mM
0.4 mL	TCEP Skin corrosion (H314)	☐ R0215	500 mM	1 mM
To 200 mL	Water, reagent-grade			

Adjust to pH 8.0 with hydrochloric acid. Add TCEP immediately before use. Store at 4 °C.

TEV elution buffer

50 mM Tris-Cl, 300 mM NaCl, 500 mM Imidazole, ☐ 1 mM TCEP, pH 8.0



DANGER



Reproductive toxicity

☐ Collect as HAZARDOUS WASTE
☐ DO NOT wash into sewer

Date: Sign: R0207

TEV storage buffer, pH 8.0, 5 ×

Amount	Ingredient		Stock	Final
62.5 mL	Tris hydrochloride (Tris-Cl), pH 8.0 R0056		1 M	125 mM
250 mL	Sodium chloride (NaCl)	R0046	5 M	2.5 M
To 500 mL	Water, reagent-grade			

Adjust to pH 8.0. For 1 × dialysis buffer, add 200 mL stock, 200 mL anhydrous glycerol, and bring to 1.0 L with water. TCEP is added directly to the dialysis bag. For glycerol diluent, combine 2 mL stock and 8 mL anhydrous glycerol; add TCEP to 0.5 mM before use. Store at 4 °C.

5 × TEV storage buffer

25 mM Tris-Cl, 500 mM NaCl, pH 8.0

[At 1 × dilution]

No GHS hazards at the indicated concentrations

- ☐ Dilute below the discharge limit
- ☐ Flush to drain with excess water

Date: Sign: R0208

TEV assay buffer, pH 8.0

Amount	Ingredient		Stock	Final
5 mL	Tris hydrochloride (Tris-Cl), pH 8.0 R0056		1 M	50 mM
10 mL	Sodium chloride (NaCl)	R0046	5 M	500 mM
333 µL	Imidazole [288-32-4] Skin corrosion (H314); Reproductive toxicity (H360)	R0027	6 M	20 mM
10 mL	Glycerol	R0022	50%	5%
200 µL	TCEP Skin corrosion (H314)	☐ R0215	500 mM	1 mM
To 100 mL	Water, reagent-grade			

Add TCEP immediately before use. Store at 4 °C. ***Note:*** For TEV standard activity test only.

TEV assay buffer

50 mM Tris-Cl, 500 mM NaCl, 20 mM Imidazole, 5% Glycerol, ☐ 1 mM TCEP, pH 8.0

DANGER

Reproductive toxicity

- ☐ Collect as HAZARDOUS WASTE
- ☐ DO NOT wash into sewer

Date: Sign: R0209

List of references

N.S. Berrow, D. Alderton, S. Sainsbury, J. Nettleship, R. Assenberg, N. Rahman, D.I. Stuart, and R.J. Owens, *Nucleic Acids Res.* **35**(6), e45 (2007).

R.B. Kapust, J. Tözsér, J.D. Fox, D.E. Anderson, S. Cherry, T.D. Copeland, and D.S. Waugh, *Protein Eng.* **14**(12), 993—1000 (2001).

Change log

2026-05-24 **Benjamin C. Buchmuller** Initial version. Consolidates the Daniel Summerer lab "Production and Activity Test of TEV Protease" (v15) preparative workflow and use notes into a single protocol. Cleavage chemistry and the buffer-compatibility windows are supplemented from the David Waugh lab "TEV Protease FAQ" (2010).

2026-10-12 **Benjamin C. Buchmuller** Activity assay recast as a five-point two-fold dilution series run in duplicate from separate aliquots, which centres the reference lot in the series and makes the lot-release criterion explicit. Imidazole ingredients tagged by CAS, so the buffer cards classify its reproductive toxicity instead of reporting an unspecified concentration.

⋮ ⋮

2026-10-19 **Benjamin C. Buchmuller** Carbenicillin replaces ampicillin. Lot release reads the freed product rather than the precursor. Cleavage conditions restated as a usable window.

2026-10-22 **Benjamin C. Buchmuller** Checked the construct figures against their primary sources: the pRK793 tag is His7 and the MBP it frees by self-cleavage is untagged at 43 kDa, and the Daniel Summerer lab sheet “Production and Activity Test of TEV Protease” (v15) gives pOPIN as 70 293 Da with $\varepsilon_{280} = 98\,570\ \mathrm{M^{-1}\,cm^{-1}}$, so its specific absorptivity is $a_{280} = 1.40\ \mathrm{L\,g^{-1}\,cm^{-1}}$ and not 1.39.

2026-10-30 **Benjamin C. Buchmuller** S219V is a slightly better catalyst than the wild type, not twice as active. The elution peak is given by its position in the gradient. Credits: Tom Muir lab, “E. coli Protein Expression Protocols”, Appendix A3.2, for the yield and the 2.0 g/L ceiling; Berrow et al. (2007) for the pOPIN 3C linker; New England Biolabs for the copper and cobalt thresholds and the pH window.

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